

Reciprocal spatial and temporal control of the H1 analysis

Response to the reviewer

We agree that the H1 comparison must address both dimensions of dependence: repeated ages within a pollen core and nearby cores within a time slice. We therefore expanded the sensitivity analysis into reciprocal temporal and spatial control while retaining human impact and climate as the substantive drivers. Time and space are structural controls, not equivalent causal competitors.

Figure 2: time within cores, space during aggregation

We first enforced one analytical row per core and 500-year age. Repeated PAP response estimates at the same age were averaged only after verifying that predictors were invariant. Conflicts and missing ages cause explicit failures. The human-plus-climate-only model is retained as a reference. The controlled model adds centred and scaled linear age as a third group. Linear age is used because cores contain at most 14 unique ages; temporal MEMs would consume too many residual degrees of freedom and require a separate detrending decision.

The main Figure 2 value remains one bounded number per core:

$$\frac{\max(I_H, 0) - \max(I_C, 0)}{\max(I_H, 0) + \max(I_C, 0)}.$$

Cores for which both contributions are non-positive are reported and excluded from this presentation profile. Continent-specific positive dbMEMs are selected from the signed time-controlled human-minus-climate difference conditional on continent-by-climate-zone strata and reused for the bounded balance. Markers are predictions at dbMEM equal to zero; distributions show observed controlled core values.

profile	original_balance	controlled_balance	min_thinning_agreement	robustness_class
signed	-0.532	-0.283	1	robust
zero_truncated	-0.512	-0.477	1	robust

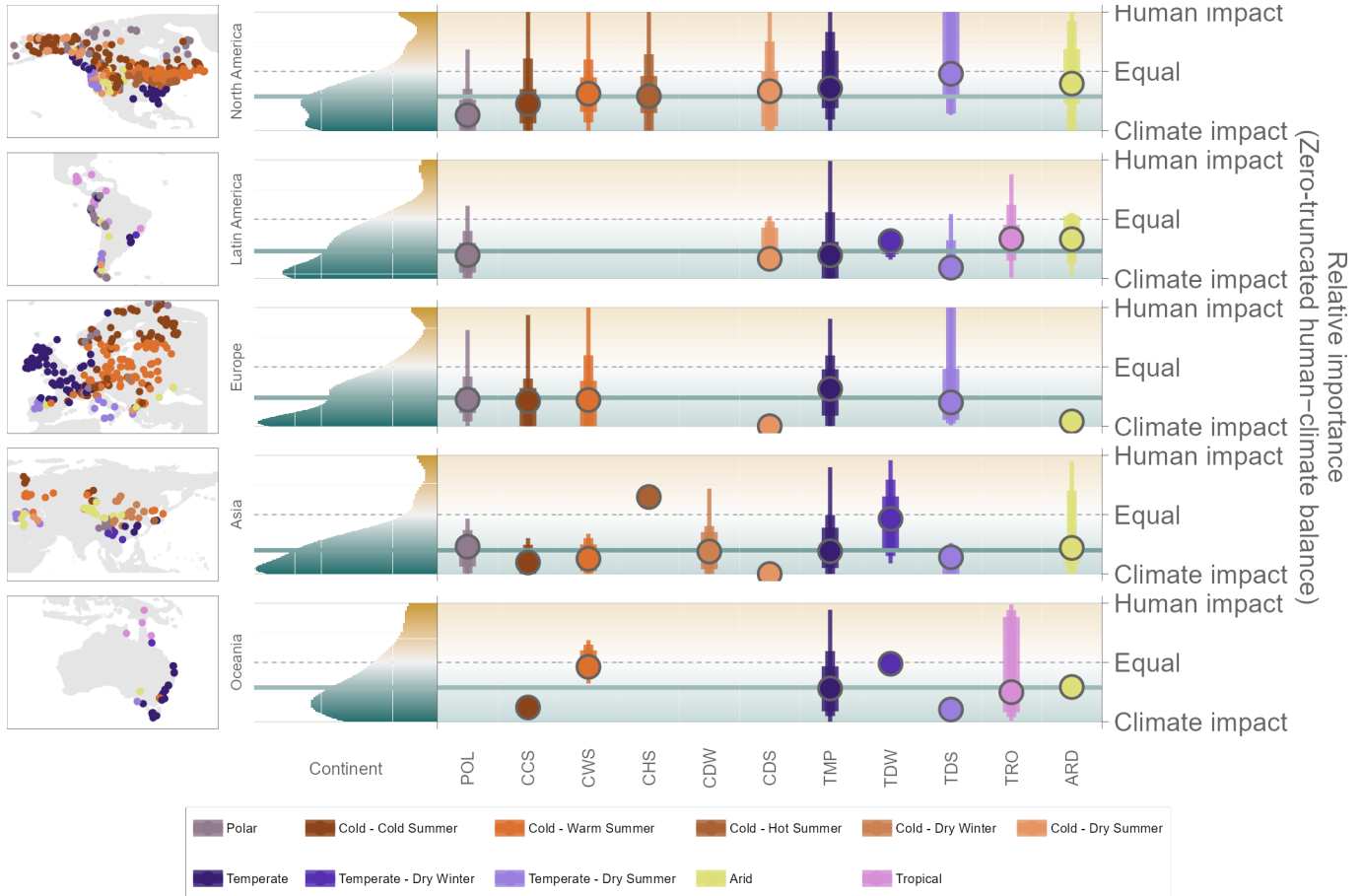


Figure 1: Time-controlled core balances and spatially filtered regional markers.

Figure 3: space within time slices

Each region-by-age model uses one row per core. Positive dbMEMs are selected against the multivariate PAP response conditional on the complete human and climate groups using a global test and Blanchet double stopping. When no conditional spatial signal is detected, human-climate-only contributions contributions are retained and space is set to exactly zero for the presentation stack. Non-estimable models remain missing.

The revised main Figure 3 uses zero-truncated hierarchical contributions for human impact, climate, and space. Signed hierarchical contributions and pure partial adjusted-R-squared fractions are exported and plotted separately; they are related diagnostics, not interchangeable estimates.

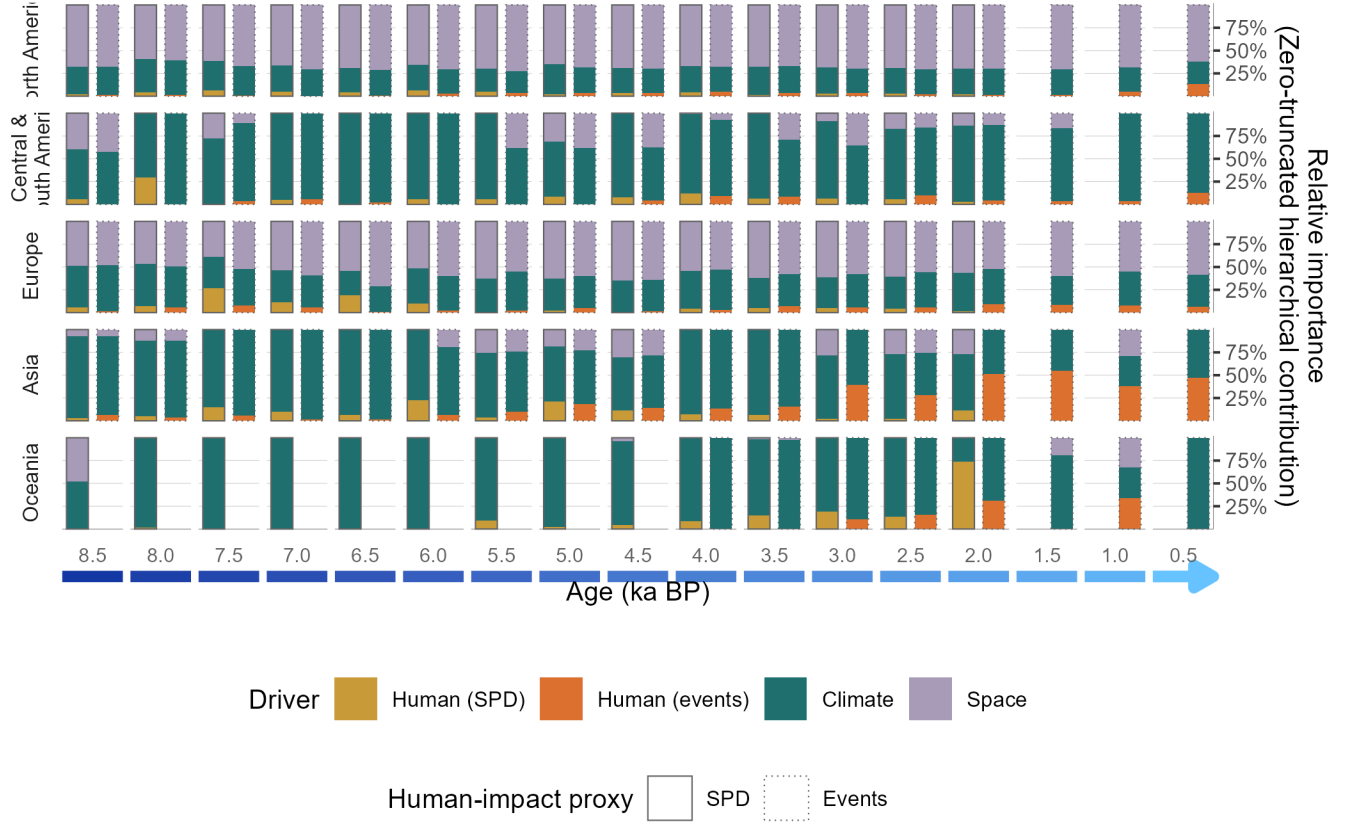


Figure 2: Human, climate, and selected spatial contributions through time.

diagnostic	count	denominator
Estimable time-controlled core-proxy models	1316	2524
Models with residual temporal dependence	7	1316
Region-age models with selected spatial terms	102	155
Models with residual spatial dependence	112	155
Estimable human-climate ranking changes	0	146

Residual temporal or spatial dependence remains visible in the diagnostics rather than being treated as automatically removed. The original two-group analyses, signed contributions,

pure fractions, actual permutation counts, selection details, thinning replicates, exclusions, and provenance are retained as supplementary machine-readable outputs. Negative adjusted fractions are never replaced with arbitrary positive values.

Interpretation boundary

The analysis tests whether the human-versus-climate conclusion is stable under explicit structural controls. It does not establish that time or space causes the PAP response, and it does not resolve finer-scale temporal autocorrelation within the original irregular pollen levels beyond the 500-year analysis units.